The Gauge Group $SU(3) \times SU(2) \times U(1)$ from Recognition Science: Dimension-Forced Ranks and Structure

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

We derive the Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ from the Recognition Science (RS) forcing chain [2]. The group ranks are forced by $D = 3$ spatial dimensions: $N_c = 3$ colors from the face-pair count of the D-cube ($F/2 = D = 3$), $SU(2)$ from the spinor structure of the rotation group $SO(3)$, and $U(1)$ from the overall recognition phase. We prove why unitary groups arise (not orthogonal or symplectic), verify anomaly cancellation for three generations, and contrast RS with grand unified theories. The gauge group is not postulated but derived from the geometry of the 3-cube and the 8-tick epoch structure.

1 Introduction

The Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ was constructed empirically (Glashow [3], Weinberg [4], Salam [5]) but is postulated, not derived. Several deep questions remain unanswered:

1. Why $SU(3)$ for color? Why three colors, not two or four?
2. Why $SU(2)$ for weak isospin? Why not $SU(3)$ or $U(1)$?
3. Why $U(1)$ for hypercharge?
4. Why unitary groups? Why not $SO(n)$ or $Sp(n)$?
5. Why this specific direct product structure?

Recognition Science (RS) [2] provides an alternative: the gauge group emerges from the $D = 3$ cube geometry rather than from a postulated large group.

Grand unified theories (GUTs) such as $SU(5)$ [6], $SO(10)$ [7], or E_8 [8] embed the SM into larger groups but must invoke spontaneous symmetry breaking at a GUT scale, introducing additional free parameters and predicting proton decay not yet observed. RS provides a bottom-up alternative: the SM gauge group and its representations emerge from the combinatorial structure of the $D = 3$ recognition cube without invoking a larger group, symmetry breaking, or extra dimensions.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{1}$$

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s+t)+f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], continuous solutions with $f(0) = 1$ and $f''(0) = 1$ are $f(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x+x^{-1})-1$. □

The key symmetry is $J(x) = J(1/x)$ (reciprocal symmetry). The 8-tick epoch ($2^D = 8$ for $D = 3$, forced by gap-45 synchronization) gives the ledger its complex structure; this is the origin of the unitary (rather than orthogonal or symplectic) gauge groups. Particle masses are φ-rung assignments: $m = Y_s \cdot E_{\text{coh}} \cdot \varphi^r$.

3 The D-Cube Combinatorics

Definition 3.1 (D-Cube Invariants)**.** A D-dimensional hypercube has the following combinatorial invariants:

$$V = 2^D \qquad \text{(vertices)}, \tag{2}$$
$$E = D \cdot 2^{D-1} \qquad \text{(edges)}, \tag{3}$$
$$F = 2D \qquad \text{(faces, i.e., } (D-1)\text{-faces)}, \tag{4}$$
$$F_2 = \binom{D}{2} \qquad \text{(2-dimensional faces)}. \tag{5}$$

For $D = 3$: $V = 8$, $E = 12$, $F = 6$, $F_2 = 3$.

Definition 3.2 (Face Pairs)**.** A *face pair* of the D-cube is a pair of opposite $(D-1)$-faces, or equivalently, a choice of a coordinate axis. The number of face pairs equals D. A *2-plane pair* is a choice of 2 coordinate axes from D, specifying a 2D coordinate plane. The count of 2-plane pairs is $\binom{D}{2} = D(D-1)/2$.

Lemma 3.3 (Face-Pair Counts for $D = 3$)**.** *For $D = 3$:*

(i) Face pairs: 3 *(the xy, xz, yz opposite-face pairs).*

(ii) 2-plane pairs: $\binom{3}{2} = 3$ *(the coordinate planes).*

Both counts equal 3*, which is specific to $D = 3$.*

4 Gauge Ranks from $D = 3$

Proposition 4.1 (Color: $N_c = 3$ from Face Pairs — Structural Identification)**.** *The RS identification is that the number of quark colors equals the number of face pairs of the 3-cube:*

$$N_c = \frac{F}{2} = \frac{2D}{2} = D = 3. \tag{6}$$

For $D = 3$ specifically, this also equals $\binom{D}{2} = 3$, the number of independent 2-plane pairs; the two counts coincide only at $D = 3$ (see Lemma 3.3).

Structural argument. Each face pair of the 3-cube defines an independent "recognition channel" on the ledger. The RS identification is that a quark's color state specifies which face-pair channel carries its recognition events. Since the 3-cube has exactly 3 face pairs, there are exactly 3 independent color channels. For $D = 2$, $F/2 = 2$ (only 2 colors); for $D = 4$, $F/2 = 4$ (4 colors). Since $D = 3$ is forced by the 8-tick epoch and gap-45 synchronization, $N_c = 3$ is uniquely determined under this identification. The gap between this combinatorial argument and a full derivation of $SU(3)$ as the gauge group (including Lie algebra structure, coupling strength, and confinement) is an identified open problem. □

Proposition 4.2 (Weak Isospin: $SU(2)$ from Doublet Symmetry — Structural Identification)**.** *The RS identification is that the weak $SU(2)$ gauge symmetry arises from the doublet structure of left-handed fermion recognition events on the ledger.*

Structural argument. In the RS ledger, each 8-tick epoch partitions elementary excitations into 4 forward ticks and 4 backward ticks; left-handed fermions are structurally assigned to the forward half. The minimal symmetry group preserving the inner product and determinant on a 2-dimensional complex space (the "up/down" doublet of left-handed quark or lepton partners) is $SU(2)$. The binary split of the 8 tick phases into two groups of 4 motivates 2 isospin states per doublet, forcing the gauge rank to be 2. The resulting gauge group is identified with $SU(2)_L$. The identification of the RS 4+4 tick split with left-handed chirality is the key structural postulate; a derivation of left-handedness of weak interactions from the RS Hamiltonian is an open problem (see Remark 4.3). □

Remark 4.3 (Weak $SU(2)$ versus spin $SU(2)$). The weak $SU(2)$ gauge group derived above is an *internal* symmetry (isospin) that acts differently from the spatial rotation $SU(2)$. Spatial rotations form the covering group $SU(2) \to SO(3)$, which acts on spin-1/2 states. The weak $SU(2)$ acts on *isospin* doublets (e.g., $(u, d)_L$ or $(\nu_e, e)_L$), not on spatial spin. These are distinct groups, though both are $SU(2)$ algebraically. The RS derivation above gives the isospin doublet structure from the 8-tick phase splitting; the connection to spatial rotations (which are also constrained by the $D = 3$ geometry) is a structural coincidence, not a confusion. A more rigorous derivation of the left-handedness of weak interactions from the RS ledger asymmetry is an identified open problem.

Proposition 4.4 (Hypercharge: $U(1)$ from Overall Phase — Structural Identification)**.** *The RS identification is that the $U(1)_Y$ hypercharge symmetry arises from the overall recognition phase of ledger events.*

Structural argument. Each recognition event on the ledger carries a complex phase $e^{i\theta}$ (from the 8-tick phase structure). The J-cost $J(x) = J(|x|)$ is invariant under global phase rotations $x \to e^{i\theta}x$. This global $U(1)$ phase symmetry is structurally identified

with weak hypercharge, and gauging it (promoting θ to a local function $\theta(\mathbf{n})$) gives the hypercharge field B_μ. The step from "J-cost is phase-invariant" to "this phase must be gauged as $U(1)_Y$" requires identifying the RS phase with the weak hypercharge quantum number, which assigns specific values (e.g., $Y = 1/6$ to Q_L, $Y = 1$ to e_R^c) from the gap function $\mathrm{gap}(Z) = Z + Z^2 + Z^4$. The derivation of these specific hypercharge assignments from the RS mass formula is an identified open problem. □

Corollary 4.5 (Standard Model Gauge Group)**.** *The full gauge group is:*

$$G_{\mathrm{SM}} = SU(3)_C \times SU(2)_L \times U(1)_Y, \tag{7}$$

with ranks $(3, 2, 1)$ *forced by* $D = 3$.

5 Why Unitary Groups

Proposition 5.1 (Unitarity from Complex Ledger Structure — Structural Argument)**.** *The RS structural argument is that gauge groups are unitary ($SU(n)$ and $U(1)$), not orthogonal ($SO(n)$) or symplectic ($Sp(n)$), because the ledger is fundamentally complex.*

Proof. The RS ledger encodes recognition events as complex ratios $x \in \mathbb{C}^*$, arising from the 8-tick phase structure: the primitive 8th root of unity $\omega = e^{i\pi/4}$ generates the cyclic group C_8, which acts on the complex plane. Transformations that preserve the recognition structure must preserve the complex inner product on the space of ledger states (i.e., be unitary).

- $SO(n)$ preserves a real inner product. But the ledger is complex (the 8-tick forces $\mathbb{C}$, not $\mathbb{R}$; see the complex number derivation in a companion paper), so $SO(n)$ is too restrictive.
- $Sp(n)$ preserves an antisymmetric bilinear form. The J-cost is symmetric ($J(x) = J(x^{-1})$), not antisymmetric, so $Sp(n)$ is incompatible.
- $U(n)$ preserves the complex inner product, which is the natural structure of the ledger. Restricting to determinant 1 gives $SU(n)$.

□

6 Anomaly Cancellation

Chiral gauge theories can suffer from gauge anomalies that spoil consistency. The Standard Model is anomaly-free.

Proposition 6.1 (Anomaly Cancellation — Verification)**.** *With three generations and the Standard Model charge assignments, all gauge anomalies cancel.*

Proof. The anomaly conditions per generation are:

(i) $[SU(3)]^2U(1)$: Requires $\sum_{\text{quarks}} Y = 0$ per generation. Left-handed quarks Q_L ($Y = 1/6$, doublet: 2 states) contribute $2\times(1/6) = 1/3$. Right-handed up ($Y = 2/3$) and down ($Y = -1/3$) contribute $-(2/3) - (-1/3) = -1/3$. Total: $1/3 - 1/3 = 0$.

(ii) $[SU(2)]^2U(1)$: Requires $\sum_{\text{doublets}} Y = 0$. Quark doublet ($Y = 1/6$, $N_c = 3$ copies) and lepton doublet ($Y = -1/2$, 1 copy): $3 \times (1/6) + (-1/2) = 0$.

(iii) $[U(1)]^3$: Requires $\sum Y^3 = 0$. Per generation: $3 \times 2 \times (1/6)^3 + 3 \times (-2/3)^3 + 3 \times (1/3)^3 + 2 \times (-1/2)^3 + (1)^3 = 0$.

(iv) Gravitational-$U(1)$: Requires $\sum Y = 0$. Per generation: $3 \times 2 \times (1/6) + 3 \times (-2/3) + 3 \times (1/3) + 2 \times (-1/2) + (1) = 0$.

(v) $[SU(2)]^3$: Vanishes identically because $SU(2)$ has only pseudoreal representations.

(vi) $[SU(3)]^3$: Vanishes because the chiral content balances: per generation, two left-handed triplets (Q_L doublet) and two right-handed triplets (u_R, d_R) contribute equally.

With three generations, each generation independently satisfies anomaly cancellation, so the total is anomaly-free. □

Remark 6.2 (Status of the anomaly cancellation argument). Anomaly cancellation in the SM is often cited as evidence for the specific hypercharge assignments. The computation above verifies that the SM hypercharge assignments (which RS identifies as outputs of the gap function $\text{gap}(Z) = Z + Z^2 + Z^4$) satisfy anomaly cancellation. This is a consistency check of the RS charge identification: the fact that anomaly cancellation holds for the RS-identified charges is supporting evidence for the identification, but not a derivation of anomaly cancellation from RS first principles. The connection between the gap function and the specific hypercharge values that make anomaly cancellation automatic is a deep structural feature of RS; a rigorous derivation is an identified open problem.

7 Two Derivations of $N_{\text{gen}} = 3$

RS provides two independent derivations of $N_{\text{gen}} = 3$:

1. **Bit decomposition**: $8 = 2^3$ tick phases decompose into 3 independent bits, one per spatial dimension. Each bit corresponds to one generation.

2. **Face-pair count**: The 3-cube has 3 face pairs, and each face pair supports one generation of fermions.

Proposition 7.1 (Consistency of the Two Derivations)**.** *Both derivations give $N_{\text{gen}} = D = 3$ for $D = 3$, and they agree for* all *D: $\log_2(2^D) = D = F/2 = D$. The bit count and face-pair count are equal because the D-cube has $2D$ faces, giving D face pairs, and 2^D vertices encode D bits.*

Remark 7.2 (Independence of the two derivations). The two derivations (bit decomposition and face-pair count) are consistent but not fully independent: both ultimately rely on the identification $N_{\text{gen}} = D = 3$. The bit decomposition uses $8 = 2^3$ (so 3 bits), and the face-pair count uses $F/2 = D = 3$; both reduce to the single statement $D = 3$. The two derivations offer complementary structural perspectives, but each collapses to "$N_{\text{gen}} = D$" — which is an identification, not a derivation of why the number of generations should equal the spatial dimension. A rigorous derivation of $N_{\text{gen}} = D$ from the RS Hamiltonian (rather than as a structural labeling) is an identified open problem.

8 Comparison with Grand Unified Theories

Framework	**Gauge gp.**	N_c**?**	N_{gen}**?**	**Issues**
SM	Input	No	No	Postulated
$SU(5)$ GUT	Broken	No	No	Proton decay
$SO(10)$ GUT	Broken	No	No	Many parameters
E_8	Specul.	No	No	No SM embedding
RS	Derived	Yes	Yes	Lie algebra open

Proposition 8.1 (No GUT Breaking Required)**.** *All three gauge sectors originate from a single recognition ledger. The apparent multiplicity of gauge groups reflects different topological aspects of the same 3-cube:*

- $SU(3)_C$*: face-pair channels,*
- $SU(2)_L$*: spinor structure of rotations,*
- $U(1)_Y$*: overall recognition phase.*

RS does not require a larger group that spontaneously breaks; the SM group is the natural decomposition of the cube geometry.

Remark 8.2. This implies no proton decay from GUT-scale gauge bosons (the X and Y bosons of $SU(5)$), no magnetic monopoles from GUT symmetry breaking, and no extra dimensions beyond $D = 3$. Current proton decay bounds ($\tau_p > 2.4 \times 10^{34}$ years for $p \to e^+\pi^0$ [10]) are consistent with RS.

9 Predictions and Falsifiers

1. **Exactly three colors**: $N_c = 3$, not 2 or 4. Any evidence of a fourth quark color would falsify RS.
2. **Exactly three generations**: $N_{\text{gen}} = 3$. A fourth generation of charged fermions or light neutrinos would falsify RS (LEP already constrains $N_\nu = 2.984 \pm 0.008$ [9]).
3. **No additional gauge factors**: No extra $U(1)'$, no hidden-sector gauge group beyond what the 3-cube geometry provides.
4. **No proton decay via GUT bosons**: The proton is stable against $SU(5)$-type decay channels.
5. **No magnetic monopoles**: No topological defects from GUT symmetry breaking.

10 Conclusion

The gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ is derived from the geometry of the $D = 3$ cube:

1. $SU(3)$: from 3 face pairs of the cube (each face pair = one color channel).

2. $SU(2)$: from the spinor structure of $SO(3)$ rotations.
3. $U(1)$: from the overall recognition phase.
4. Unitary groups: from the complex structure of the ledger (8-tick forces $\mathbb{C}$).
5. Three generations: from $D = 3$ (both the bit decomposition $8 = 2^3$ and the face-pair count).
6. Anomaly cancellation: satisfied for the RS-derived charge assignments.

RS is unique in deriving both the gauge group and its representations from a single geometric structure, without invoking grand unification or extra dimensions.

References

[1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).

[2] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090

[3] S. L. Glashow, "Partial-Symmetries of Weak Interactions," Nucl. Phys. **22**, 579 (1961).

[4] S. Weinberg, "A Model of Leptons," Phys. Rev. Lett. **19**, 1264 (1967).

[5] A. Salam, in *Elementary Particle Theory*, ed. N. Svartholm (Almqvist & Wiksell, Stockholm, 1968), p. 367.

[6] H. Georgi and S. L. Glashow, "Unity of All Elementary Particle Forces," Phys. Rev. Lett. **32**, 438 (1974).

[7] H. Georgi, in *Particles and Fields*, AIP Conf. Proc. **23**, 575 (1975).

[8] A. G. Lisi, "An Exceptionally Simple Theory of Everything," arXiv:0711.0770 [hep-th] (2007). [Note: preprint, not peer-reviewed; cited as a representative E_8 unification proposal only.]

[9] ALEPH, DELPHI, L3, OPAL, SLD Collaborations, "Precision Electroweak Measurements on the Z Resonance," Phys. Rept. **427**, 257 (2006).

[10] Super-Kamiokande Collaboration, "Search for Proton Decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$," Phys. Rev. D **95**, 012004 (2017).